

Snort RAG PDF Upload Test - Web Attack KB Extension

Project-specific document for testing retrieval enrichment for SQL injection and path traversal.

Document role	Add compact SQL injection and directory traversal examples for retrieval and rule-generation testing.
Project	Snort RAG Rule Generator - NLP + Retrieval-Augmented Generation for defensive Snort rule generation.
Use in dashboard	Upload this PDF in the PDF Upload tab to test temporary in-memory knowledge-base extension.
Version	Version 2 - SQLi and Traversal KB
Generated	2026-06-09

1. Attack families: SQL injection and path traversal

This document gives compact project-specific examples for two web-application attack families. It is designed to test whether the PDF upload mechanism can enrich retrieval with additional Snort rule patterns.

- SQL injection label: `sql_injection`. Main indicators include `UNION SELECT`, `OR 1=1`, `information_schema`, `sleep()`, `benchmark()`, and SQL comments such as `--`.
- Path traversal label: `directory_traversal`. Main indicators include `../`, `..\`, `%2e%2e%2f`, `/etc/passwd`, `boot.ini`, and encoded traversal payloads.
- Both families usually occur in HTTP URI parameters or HTTP request bodies.
- The rule should be defensive, specific, and documented with an explanation.

2. Retrieval-ready examples

Attack type	Description	Useful Snort content
<code>sql_injection</code>	Detect SQL injection with <code>UNION SELECT</code> in HTTP URI.	<code>union select, nocase</code>
<code>sql_injection</code>	Detect SQL boolean bypass using <code>OR 1=1</code> in a login request.	<code>or 1=1, login</code>
<code>directory_traversal</code>	Detect path traversal trying to read <code>/etc/passwd</code> .	<code>../, /etc/passwd</code>
<code>directory_traversal</code>	Detect encoded traversal using <code>%2e%2e%2f</code> .	<code>%2e%2e%2f, nocase</code>

3. Example SQL injection rule

```
alert tcp $EXTERNAL_NET any -> $HOME_NET $HTTP_PORTS (msg:"KB SQL injection UNION SELECT in URI"; flow:to_server,established)
```

4. Example path traversal rule

```
alert tcp $EXTERNAL_NET any -> $HOME_NET $HTTP_PORTS (msg:"KB directory traversal passwd access"; flow:to_server,established)
```

5. False-positive reduction notes

- Avoid triggering only on the word `select` when it appears in normal documentation, product text, or a CSS selector explanation.

- Avoid a broad traversal rule that alerts on every two dots in a URL. Combine traversal markers with sensitive file names or encoded traversal signatures.
- Prefer multiple content conditions or context-specific HTTP modifiers when the rule would otherwise be too broad.
- Record improvement suggestions in the dashboard when missing options or broad content patterns increase false-positive risk.